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Portfolio Analysis Using Fama-Macbeth Regression

Introduction:

Fama-Macbeth procedure:

The Fama-MacBeth 1973 procedure is a two-step regression method primarily used in finance to empirically test asset pricing models like the Capital Asset Pricing Model (CAPM).

It involves running monthly cross-sectional regressions of the form:

$$R_{pt} = \hat{\gamma}_{0t} + \hat{\gamma}_{1t} \cdot \hat{\beta}_{p,t} + \hat{\gamma}_{2t} \cdot \hat{\beta}_{p,t-1}^2 + \hat{\gamma}_{3t} \cdot \bar{s}_{p,t} + \hat{\eta}_{p,t}$$

Where $p = 1, \dots, n_p$ the portfolios and t are the time periods (months).

The Fama- Macbeth in general involves the following steps.

Step 1: Cross-Sectional Regressions

Our objective is to estimate the relationship between returns and risk factors for each period separately. For each time-period (month), we run a regression of the returns of portfolios on the risk factors predicted by the CAPM. Then we obtain a series of risk premiums (coefficients on the risk factors) for each period, which are estimates of how much return is associated with each unit of risk.

Step 2: Time-Series Analysis

Our objective is to determine the average risk premium over time and test its significance. We calculate the average of the estimated risk premiums from the first step over all periods. Then we conduct statistical tests to assess if these average premiums are significantly different from zero. This provides evidence on whether the risk factors (like the market premium in CAPM) are consistently priced in the market and whether the CAPM holds empirically.

The above is a generalization of the procedure which we will use more specifically to empirically test the context of the CAPM in our provided data.

By applying the Fama- Macbeth to CAPM, we can empirically test one of the model's core predictions: that the expected excess return on an asset is proportional to its market beta. The Fama-MacBeth method provides a way to estimate and test the risk premium associated with market risk. If the procedure finds that the average risk premium for market risk is significantly positive and other factors do not significantly explain returns, this supports CAPM's assertion that market risk is the primary determinant of asset returns.

The Fama-MacBeth procedure is particularly useful for this kind of testing because it accounts for potential issues like autocorrelation and heteroskedasticity in the error terms of financial data, providing more reliable estimates of the relationship between returns and risk as proposed by asset pricing models like CAPM.

Data analysis and discussion:

The table below, presents the sample estimates obtained from the Fama – Macbeth procedure.

Sample Estimates	$\widehat{\gamma}_0$	$\widehat{\gamma}_1$	$\widehat{\gamma}_2$	$\widehat{\gamma}_3$	$r_m - r_0$	r_{ft}
1980-1986 1987-1991 1992-1995	3.24	-6.98	4.29	-0.06	-2.4	0.34
1984-1990 1991-1995 1996-1999	0.72	0.197	1.1	-0.14	1.37	0.406
1988-1994 1995-1999 2000-2003	0.552	1.261	-1.55	0.024	-0.498	0.246
1992-1998 1999-2003 2004-2007	-0.83	1.45	-1.41	0.19	1.43	0.28
1996-2002 2003-2007 2008-2011	0.82	0.42	-0.043	-0.141	-0.28	0.035
2000-2006 2007-2011 2012-2015	1.66	-2.06	0.93	0.004	-0.56	0.005

The sample estimates derived from the Fama-MacBeth regression, highlight key coefficients ($\gamma_0, \gamma_1, \gamma_2, \gamma_3$), market risk premiums ($r_m - r_0$), and risk-free rates (r_{ft}) across various time periods. The intercept (γ_0), representing the Sharpe-Lintner hypothesis, fluctuates notably, with values such as 3.24 in 1980-1986 and -0.83 in 1992-1998, suggesting

potential mispricing or omitted factors affecting returns. Similarly, the market beta coefficient (γ_1) shows inconsistencies, ranging from -6.98 in 1980-1986 to 1.45 in 1999-2003, challenging CAPM's assertion of a consistent positive risk-return relationship. The linearity coefficient (γ_2) varies across periods, peaking at 4.29 in 1980-1986 and dropping to -1.41 in 1999-2003, indicating non-linear risk-return dynamics during certain periods. The non-systematic risk coefficient (γ_3) remains close to zero in most cases, such as -0.141 in 1996-2002, supporting the CAPM prediction that non-systematic risks are not systematically priced.

Additionally, the market risk premium ($r_m - r_0$) demonstrates significant variability, with positive values like 1.43 in 1992-1998 and negative values like -2.4 in 1980-1986, reflecting inconsistencies in investor compensation for market risk. The risk-free rate (r_{ft}) also fluctuates, peaking at 0.406 in 1984-1990 and declining to 0.005 in 2012-2015, which further influences the dynamics of the risk-return trade off. Overall, the findings underscore deviations from CAPM's predictions in specific periods, highlighting the limitations of the model in consistently capturing the complexities of financial markets. These results suggest the importance of considering additional risk factors or alternative models to better explain asset returns across varying market conditions.

Linearity (γ_2)	t-statistic	p-value	Confidence interval	Reject/Accept CAPM
1980-1986 1987-1991 1992-1995	2.22	0.03	0.4, 8.2	Reject
1984-1990 1991-1995 1996-1999	0.72	0.48	-1.98, 4.18	Accept
1988-1994 1995-1999 2000-2003	-0.84	0.41	-5.27, 2.17	Accept
1992-1998 1999-2003 2004-2007	-1.62	0.11	-3.15, 0.34	Accept
1996-2002 2003-2007 2008-2011	-0.038	0.9697	-2.32, 2.23	Accept
2000-2006 2007-2011 2012-2015	0.78	0.44	-1.46, 3.32	Accept

Testing the Linearity checks if the relationship between variables (e.g., stock returns and risk factors) is linear, meaning a change in one variable results in a proportional change in another. In period 1, the t-statistic is 2.2166 with a p-value of 0.03152, indicating that we can reject the null hypothesis of the expected value of γ_{2t} being zero at the 5% significance level. In this case, a significant γ_{2t} would imply a non-linear risk-return relationship, which is inconsistent with the predictions of CAPM. Similarly, we can argue for the results of each of the other time periods and why we reject or accept the CAPM.

No systematic effects of non- β risk (γ_3)	t-statistic	p-value	Confidence interval	Reject/Accept CAPM
1980-1986 1987-1991 1992-1995	-0.483	0.631	-0.324, 0.198	Accept
1984-1990 1991-1995 1996-1999	-1.05	0.299	-0.41, 0.13	Accept
1988-1994 1995-1999 2000-2003	0.15	0.88	-0.29, 0.34	Accept
1992-1998 1999-2003 2004-2007	1.4	0.169	-0.083, 0.46	Accept
1996-2002 2003-2007 2008-2011	-1.09	0.28	-0.403, 0.12	Accept
2000-2006 2007-2011 2012-2015	0.03	0.98	-0.23, 0.24	Accept

Non-Systematic Effects of non- β risk evaluates the impact of factors unique to a particular asset or market that do not affect the market, such as company-specific news. For period 1, the t-statistic is -0.48318 with a p-value of 0.6312, indicating that we cannot reject the null hypothesis of $E[\gamma_3]=0$ at conventional significance levels, including 1% and 5%. This suggests that there is insufficient evidence to conclude that there is a systematic non-beta risk effect during this testing period. It would be consistent with the CAPM prediction that non-market risk does not have a systematic effect on asset returns. With the same thought process, we can derive the results for the other periods.

Positive expected return-risk trade off (γ_1)	t-statistic	p-value	Confidence interval	Reject/Accept CAPM
1980-1986 1987-1991 1992-1995	-1.17	0.25	-12.43, 3.25	Accept
1984-1990 1991-1995 1996-1999	-0.465	0.643	-6.196, 3.852	Accept
1988-1994 1995-1999 2000-2003	0.615	0.54	-3.95, 7.47	Accept
1992-1998 1999-2003 2004-2007	0.009	0.99	-3.57, 3.61	Accept
1996-2002 2003-2007 2008-2011	0.224	0.823	-5.54, 6.95	Accept
2000-2006 2007-2011 2012-2015	-0.467	0.642	-7.87, 4.89	Accept

Positive Expected Return Trade-off investigates the assumption that higher risk (volatility) should be compensated by higher expected returns, as per traditional finance theory. We filter for the first firm permno=10145 and extract the vector of values for the market return. The t-statistic is -1.1707 with a p-value of 0.2465, indicating that the null hypothesis, which states there is no significant difference in the means of γ_1 and the excess returns over a risk-free asset, cannot be rejected at the conventional 5% or even at the 10% significance levels. This suggests that there is no evidence of a positive expected return-risk trade off, which is a deviation from the CAPM expectations. With the explanation method we can interpret the results for the other time periods.

Sharpe-Lintner (S-L) Hypothesis (γ_0)	t-statistic	p-value	Confidence interval	Reject/Accept CAPM
1980-1986 1987-1991 1992-1995	2.21	0.03	0.26, 5.54	Reject
1984-1990 1991-1995 1996-1999	0.3	0.765	-1.79, 2.42	Accept
1988-1994 1995-1999 2000-2003	0.235	0.815	-2.31, 2.93	Accept
1992-1998 1999-2003 2004-2007	-0.92	0.36	-3.52, 1.3	Accept
1996-2002 2003-2007 2008-2011	0.578	0.5658	-1.94, 3.51	Accept
2000-2006 2007-2011 2012-2015	1.39	0.17	-0.73, 4.05	Accept

Sharpe-Linder Hypothesis explores the idea that there's a direct relationship between the slope of the security market line (a representation of the Capital Asset Pricing Model) and the level of market integration, suggesting that more integrated markets will exhibit a steeper slope. The t-statistic is 2.2144 and the p-value is 0.03168, which suggests that the null hypothesis of no difference between the mean intercept and mean risk-free return can be rejected at both the 5% and 10% significance levels. This indicates that the Sharpe-Lintner hypothesis does not likely hold, thus rejecting the CAPM. Similarly, we can explain the results of each of the other time periods and why we reject or accept the CAPM.

Capital market efficiency	γ_1 (p-value, reject/accept CAPM)	γ_2	γ_3
1980-1986 1987-1991 1992-1995	0.84, Accept CAPM	0.8, Accept CAPM	0.65, Accept CAPM
1984-1990 1991-1995 1996-1999	0.24, Accept	0.14, Accept	0.197, Accept
1988-1994 1995-1999 2000-2003	0.172, Accept	0.258, Accept	0.333, Accept
1992-1998 1999-2003 2004-2007	0.739, Accept	0.958, Accept	0.715, Accept
1996-2002 2003-2007 2008-2011	0.056, Accept	0.086, Accept	0.75, Accept
2000-2006 2007-2011 2012-2015	0.99, Accept	0.989, Accept	0.098, Accept

Capital Market Efficiency assesses the market's ability to accurately reflect all available information in the prices of securities, implying that it's difficult to consistently achieve higher returns without taking on additional risk. For the γ_{1t} coefficient, the calculated Ljung-Box statistic is 0.037153 with a p-value of 0.8472, which does not lead us to reject the null hypothesis of zero autocorrelation. Similarly, for γ_{2t} , the statistic is 0.063252 with a p-value of 0.8014, and for γ_{3t} , the statistic is 0.20364 with a p-value of 0.6518. In all these cases, the null hypothesis of zero autocorrelation stands, and we conclude that the coefficients γ_{1t} , γ_{2t} , and γ_{3t} are likely independent over time.

During the first testing period of 1992-1995, the evidence does not compel us to reject the null hypothesis for any of the coefficients, suggesting that the CAPM assumptions hold for this timeframe. Keeping the same thought process in mind, we interpret the results for the other

time periods similarly and that is how we conclude whether to accept or reject the CAPM. The capital market efficiency results show that we always accept in the 6 time periods.

Conclusion:

The application of the Fama-MacBeth regression procedure provided valuable insights into the validity of the Capital Asset Pricing Model (CAPM) across various time periods. The findings reveal mixed support for CAPM's core assumptions and implications. While certain periods exhibited a positive risk-return trade-off, consistent with CAPM predictions, this relationship was not uniformly observed. The inconsistency suggests that market beta alone may not sufficiently explain the variation in expected returns, highlighting the potential influence of other factors.

The analysis found no significant systematic effects of non-beta risk, aligning with CAPM's assertion that only market risk influences asset returns. This outcome provides partial support for the model's fundamental hypothesis. However, significant non-linear relationships between risk and return were detected in some periods, contradicting the linearity assumption of CAPM. These deviations suggest the presence of omitted factors or market anomalies affecting returns during specific periods.

The evaluation of the Sharpe-Lintner hypothesis yielded varied results, with some periods supporting the hypothesis while others did not. This variability indicates changes in market integration and the behaviour of the security market line over time. On a positive note, the findings consistently supported the notion of capital market efficiency, as evidenced by the lack of autocorrelation in the residuals of the Fama-MacBeth regressions. This suggests that markets effectively reflect available information in asset prices.

Overall, the project underscores the limitations of CAPM as a universal model for explaining asset returns. While it captures certain risk-return dynamics, its explanatory power is weakened by non-linear effects and temporal variations. These results suggest that augmenting CAPM with additional risk factors or employing alternative asset pricing models may provide a more accurate representation of market behaviour. Future research could expand this analysis by incorporating multi-factor models, such as the Fama-French three-factor model, to better understand the determinants of asset returns. Additionally, exploring the impact of macroeconomic and behavioural variables could further elucidate the observed deviations from CAPM predictions.